



Universidad Nacional Mayor de
SAN MARCOS
Universidad del Perú. Decana de América



Genómica y Bioinformática
para la Biodiversidad

Generating evidence to understand the Algarrobo (*Neltuma pallida*) decline

Omic sciences and bioinformatics at the service of conservation

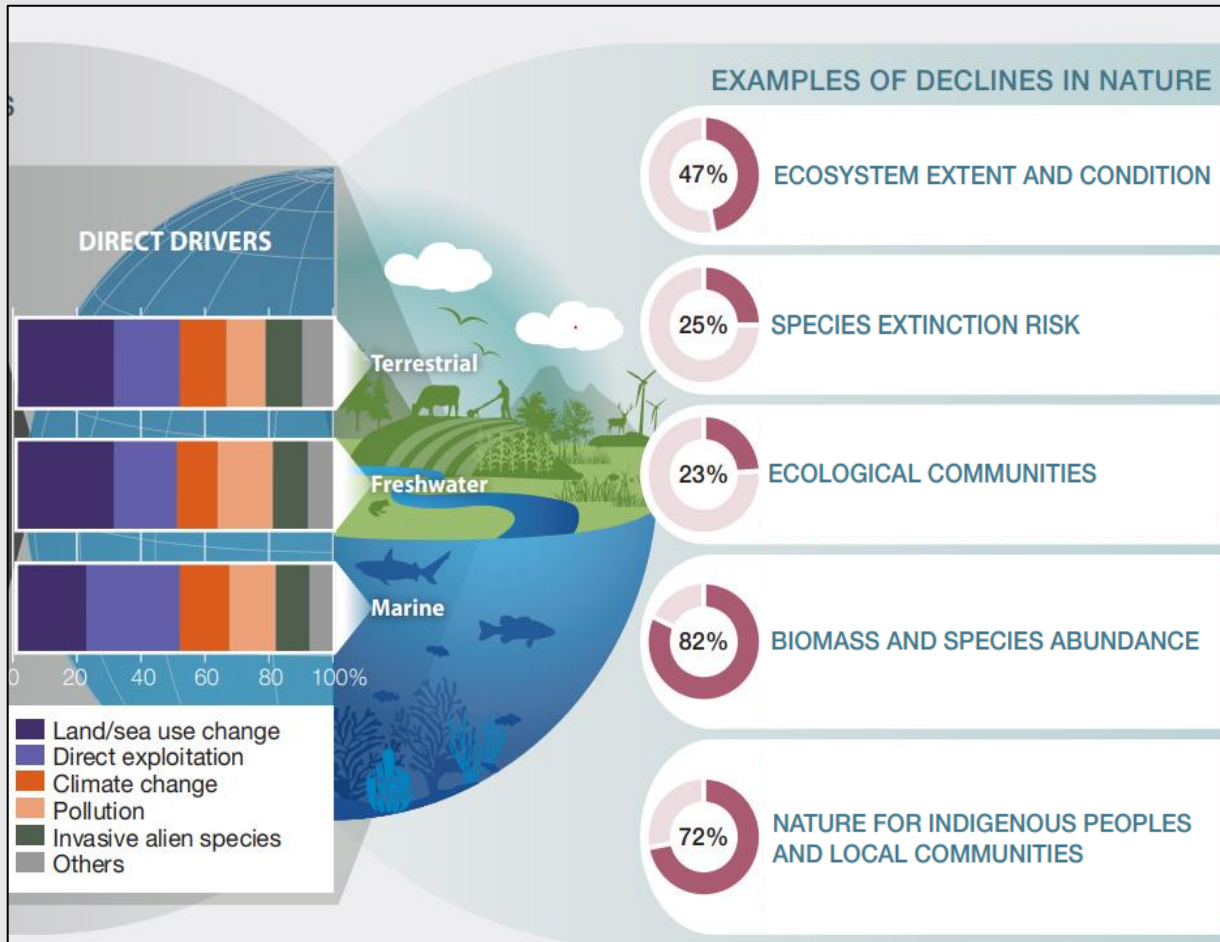
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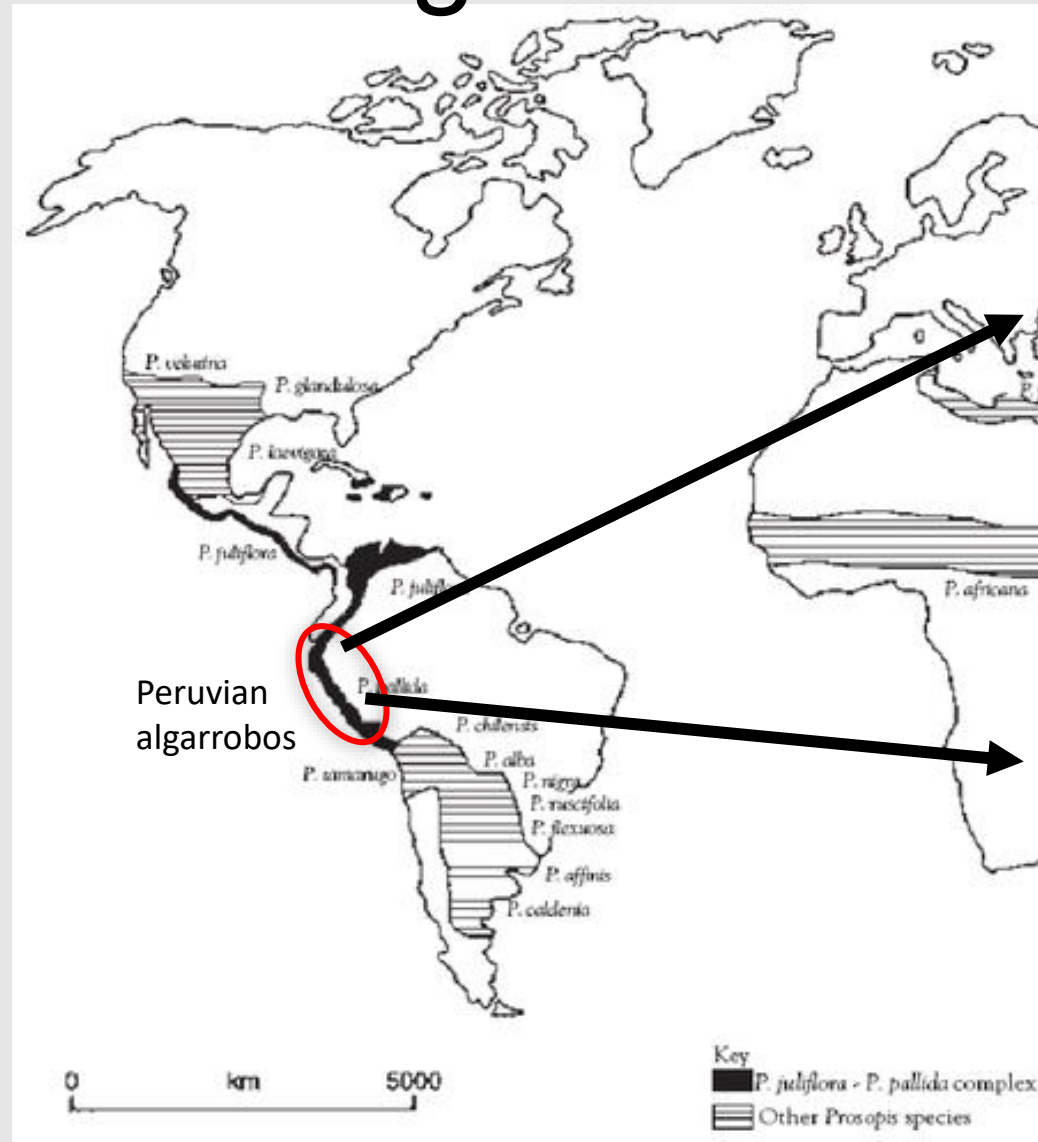
Biodiversity decline



- We are facing unprecedented biodiversity decline rates.
- Ecosystem functions and services are deteriorating.

Peruvian algarrobos

North: *N. pallida* (algarrobo)

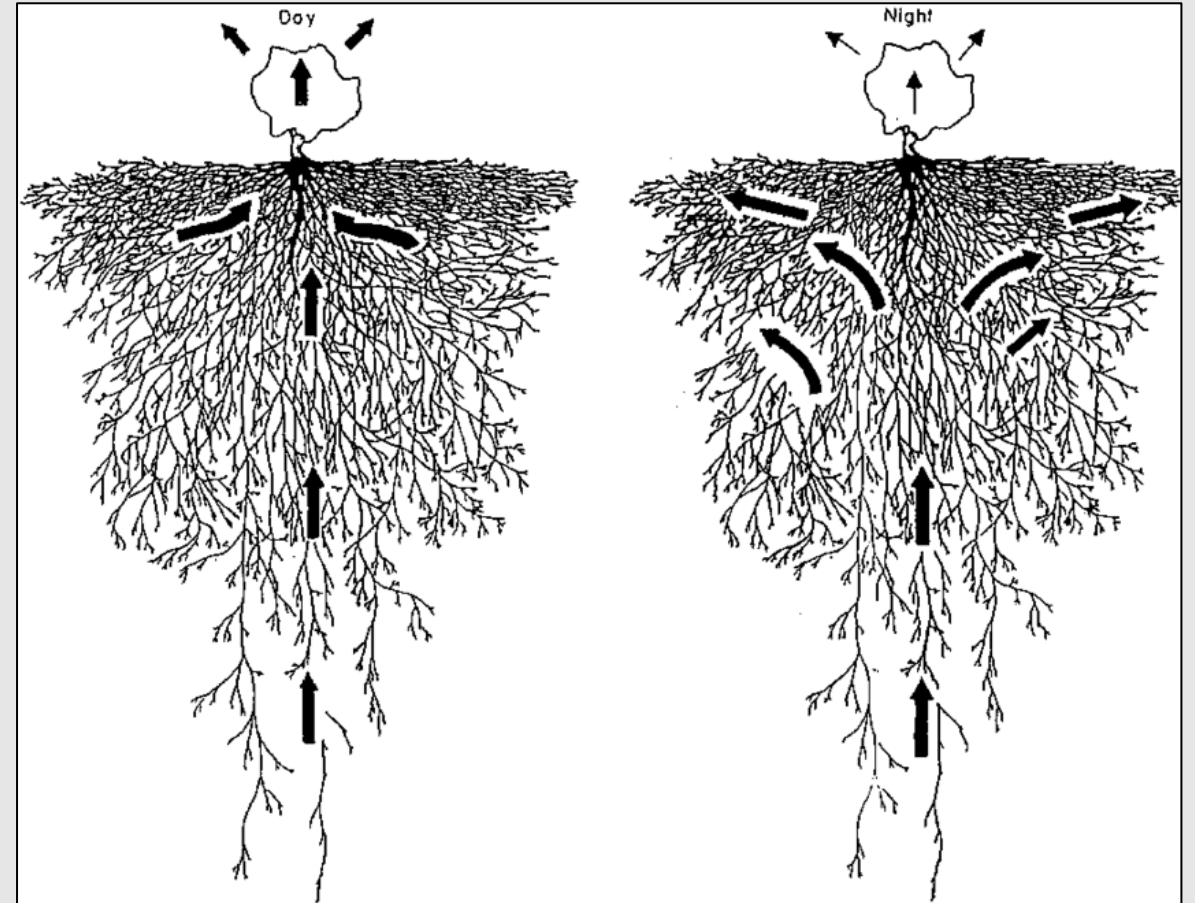


South: *N. limensis* (huarango)

Pacsieznik et al. (2001). Native distribution of *Neltuma* (ex. *Prosopis* sect. *Algarobia*)

Keystone species for dry forests

- While thriving in seasonally dry forests (even under hyper-arid conditions), Peruvian algarrobos promote the survival of other organisms.



Caldwell et al. (1998). Hydraulic lift: Consequenced of Water Efflux from the Roots of Plants. *Oecologia* 113(2)

Collapsing seasonally dry forests

- Deforestation linked to agricultural development led to desertification. Strong evidence to ancient ecological imbalances in the past.
- Current threats include overexploitation and habitat degradation.



La Lupa (2020). Huarango forest burning in Orovilca, Ica



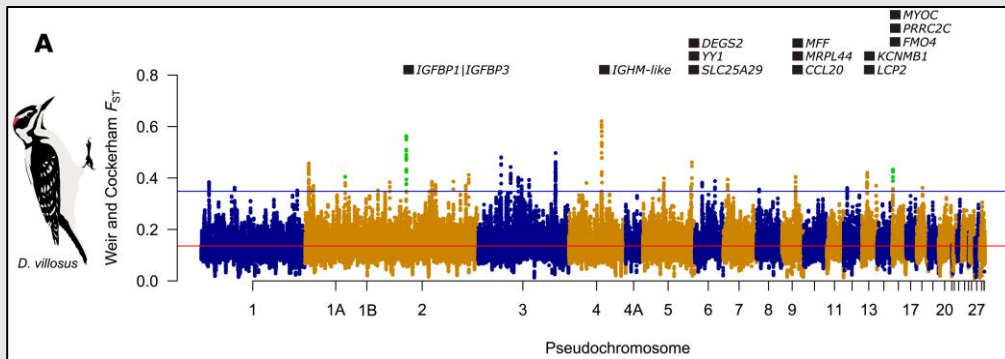
Whaley (2020). Ecology and diagnosis of *Enallodiplosis discordis*

Footprints of evolution

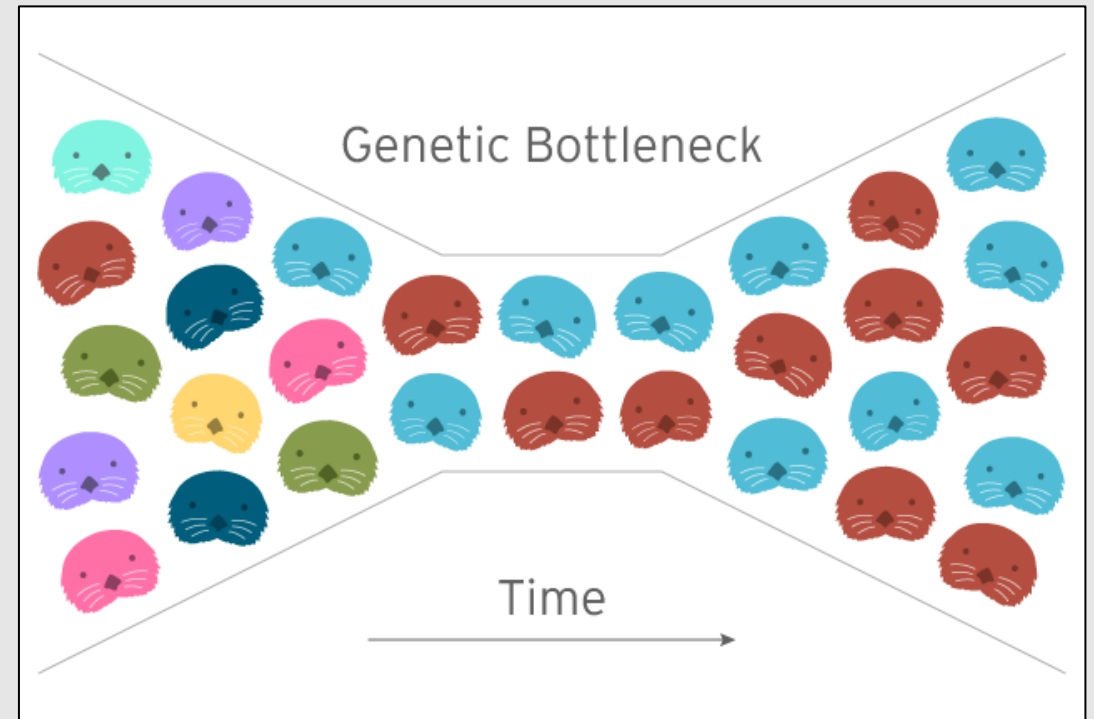
- Adaptive and neutral evolutionary processes have consequences for populations and their genomes.



Rahman (2022). Calculating F_{ST} for haploid data in R



Moreira & Smith (2023). Convergent genomic signatures of local adaptation. *Sci. Adv.* 9, add0560

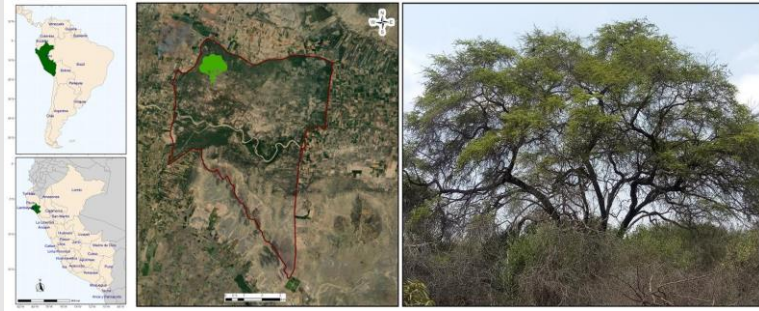


Monterey Bay Aquarium (2016). Decoding the sea otter genome

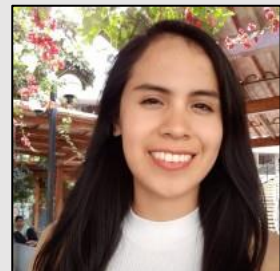
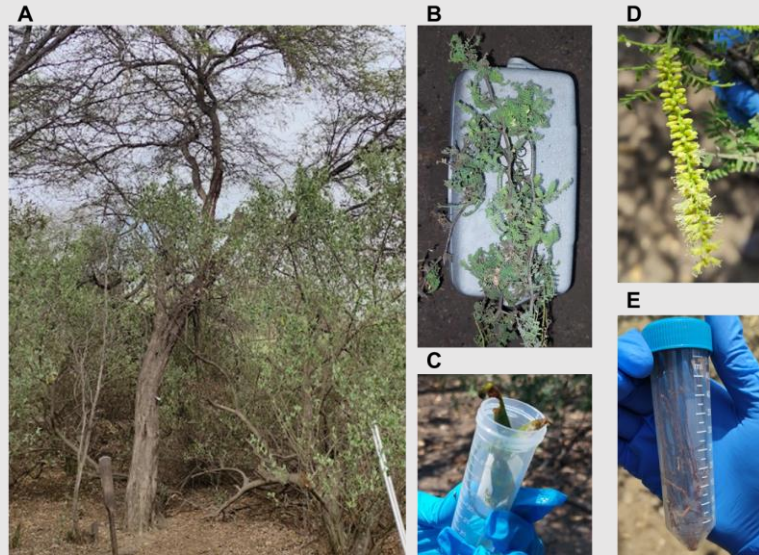
Key questions

- ¿What do we need to track these **footprints of evolution**?
- ¿What natural and anthropogenic processes shaped algarrobo's **genetic diversity**?
- ¿What is their current **genetic status**?
- ¿What kind of factors are behind its **population decline**?

Starting on an emblematic forest: Pomac



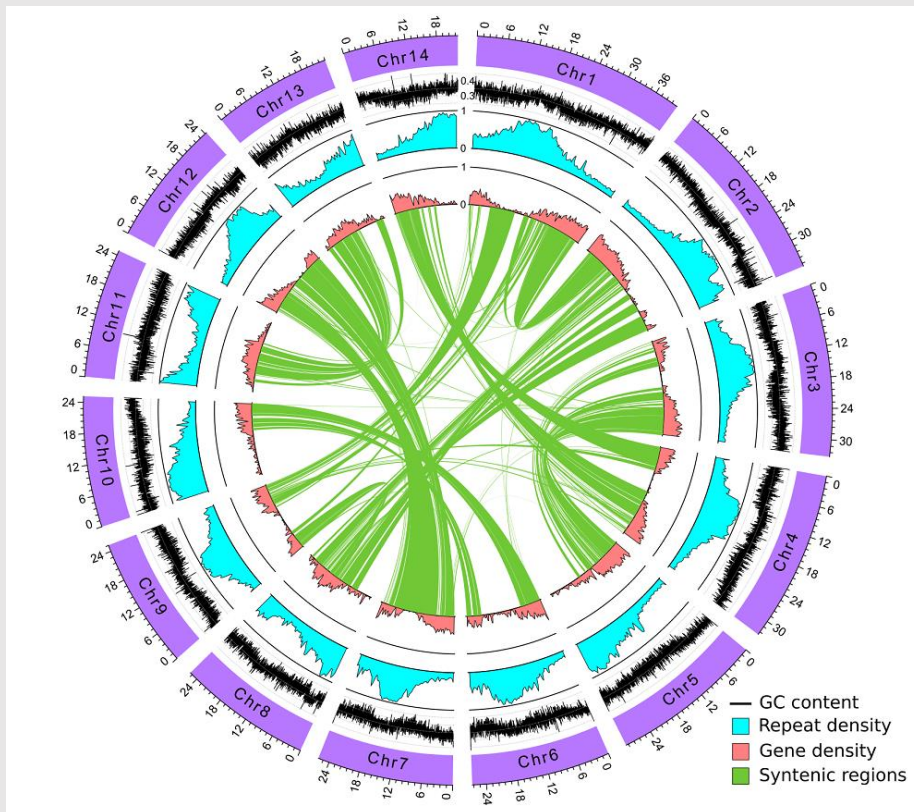
- **Genomics (chloroplast and nuclear genomes, 1 emblematic tree).**
- Population genomics (> 100 contemporary trees).
- Transcriptomic atlas (1 emblematic tree).
- Metabarcoding (rhizosphere, soil, plant tissues, gall midge).
- Palaeogenomics (200 historical and ancient trees ~ 2,000 years old).



Genomics of *N. pallida*

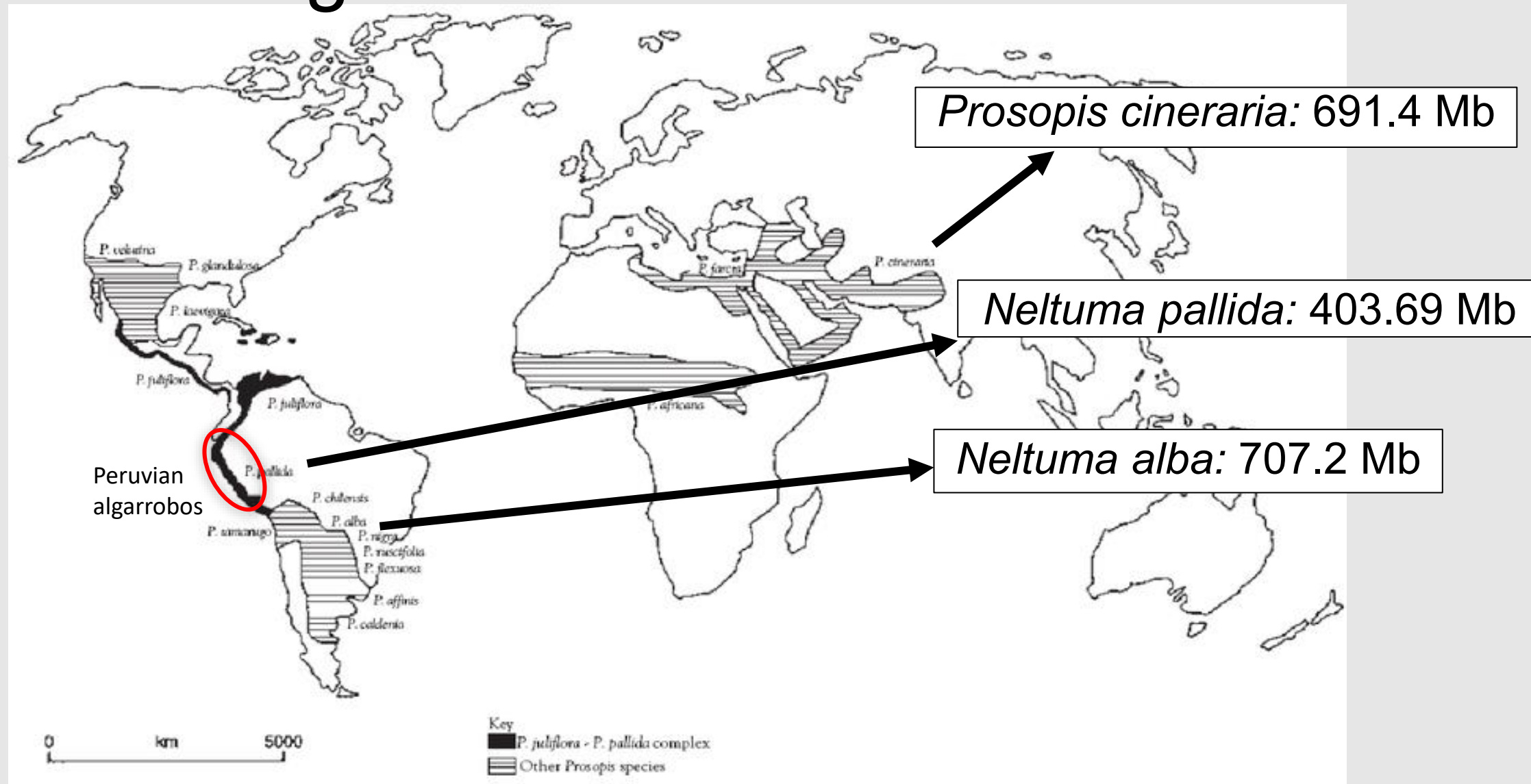
A chromosome-level genome assembly of the Peruvian Algarrobo (*Neltuma pallida*) provides insights on its adaptation to its unique ecological niche

Renato La Torre ¹, John P. Hamilton ^{2,3}, Manuel Saucedo-Bazalar ⁴, Esteban Caycho ¹, Brieanne Vaillancourt ²,
Joshua C. Wood ², Manuel Ramírez ¹, C. Robin Buell ^{2,3,5,6}, Gisella Orjeda ^{1,*}



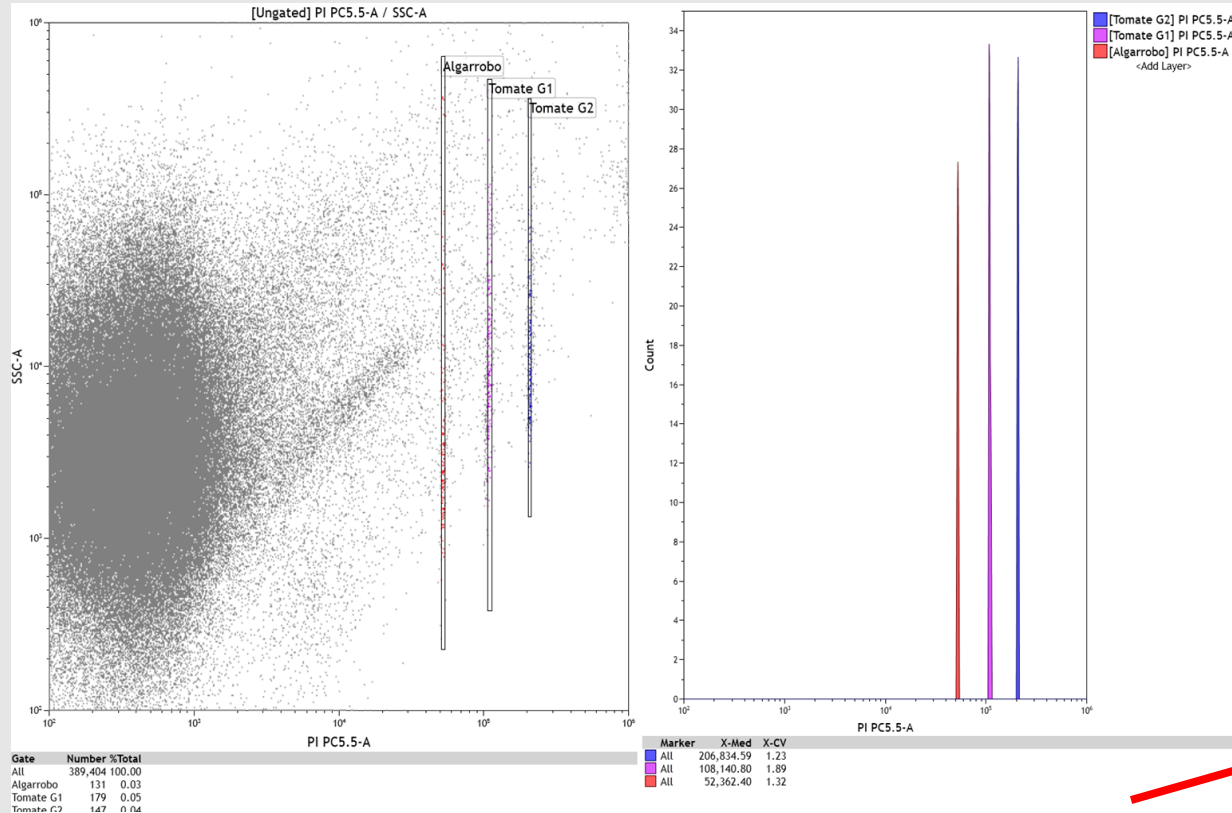
- Using an hybrid strategy for assembly (Illumina + Nanopore) and Hi-C for scaffolding.
- 14 pseudochromosomes, diploid, ~400 Mb in length.
- BUSCO completeness 99.3%. Duplicated: 3.3%.
- Transcriptome-based + public sequences + predictions annotation (MAKER).

Algarrobos' genome size



Pacsieznik et al. (2001). Native distribution of *Neltuma* (ex. *Prosopis* sect. *Algarobia*)

Flow cytometry



$$2C \text{ de } N. \text{ pallida (pg)} = 1.96 \text{ pg} \times \frac{52,387.01}{108,140.80} \approx 0.95 \text{ pg}$$

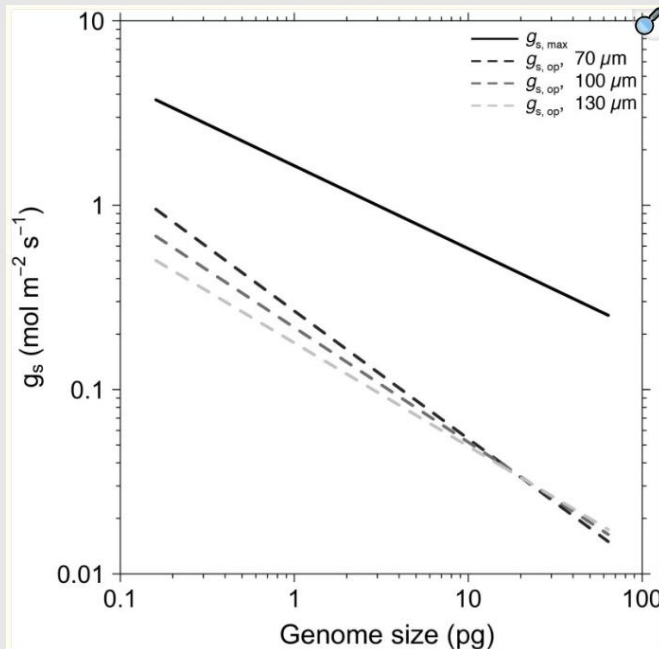
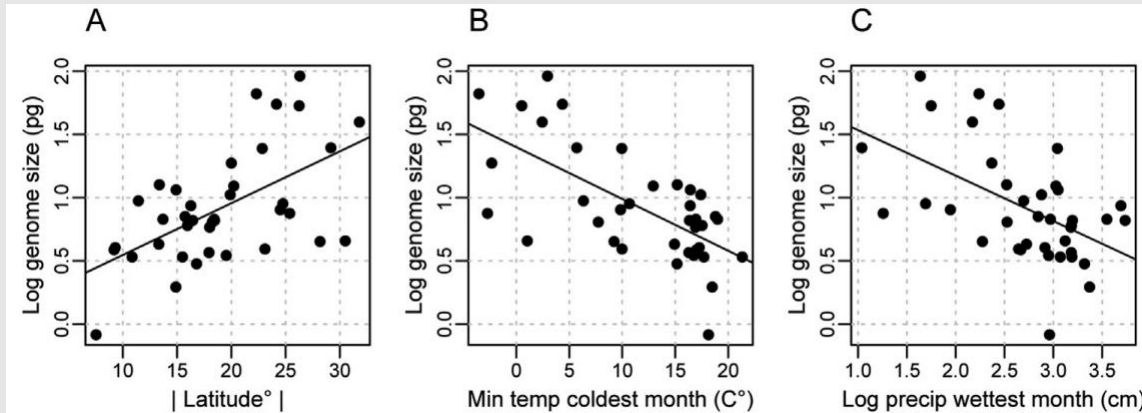
$$0.475 \text{ pg} \times 978 \frac{\text{Mbp}}{\text{pg}} = 464.55 \text{ Mbp}$$

The Genus Prosopis

<i>P. chilensis</i> (MOLINA) STUNTZ.	W. Sudan	56	1.725 ± 0.041	GHI
<i>P. chilensis</i> (MOLINA) STUNTZ.	E. Sudan	56	1.732 ± 0.046	GH
<i>P. chilensis</i> (MOLINA) STUNTZ.	Kenya	56	1.712 ± 0.017	HIJ
<i>P. chilensis</i> (MOLINA) STUNTZ.	C. Sudan	56	1.725 ± 0.037	GHI
<i>P. chilensis</i> (MOLINA) STUNTZ.	Chile	28	0.853 ± 0.024	R H
<i>P. alba</i> GRISEB.	Chile	28	0.840 ± 0.016	R H
<i>P. flexusa</i> DC.	Chile	28	0.811 ± 0.016	R H
<i>P. pallida</i> WILLD.	Peru	28	0.836 ± 0.032	R H
<i>P. glandulosa</i> TORR.	Mexico	28	0.827 ± 0.005	R H
<i>P. juliflora</i> (SWARTZ) DC.	Senegal	28	0.852 ± 0.010	R H
<i>P. juliflora</i> (SWARTZ) DC.	Mexico	28	0.852 ± 0.010	R H
<i>P. lampa</i> WILLD.	Chile	28	0.859 ± 0.012	R H
<i>P. siliquas</i> (WILLD.)	Chile	28	0.984 ± 0.026	R H

- Genome size was corroborated using Flow cytometry and previous reports (e.g., Bukhari, 1997).

Plant genome size variation

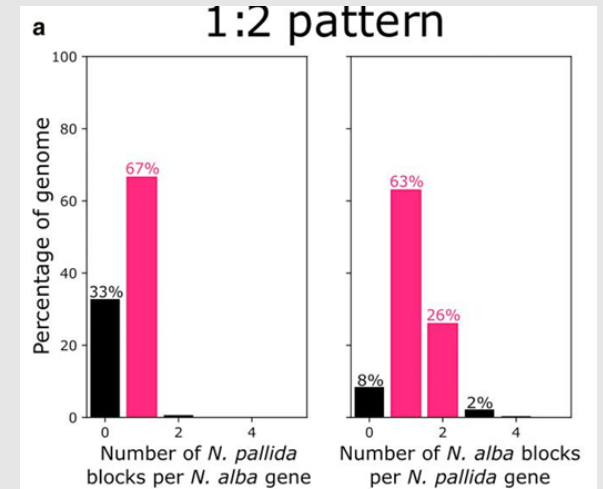
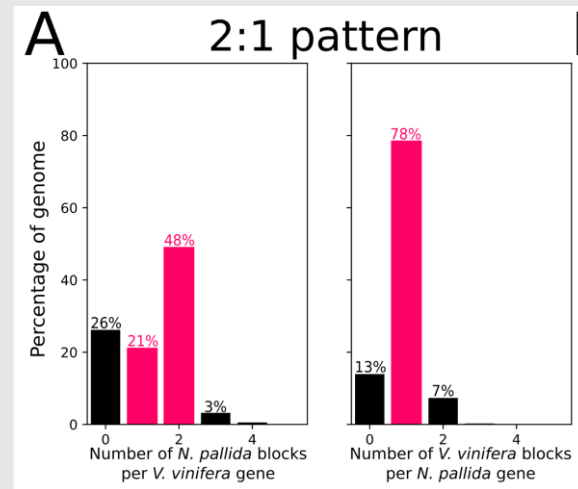
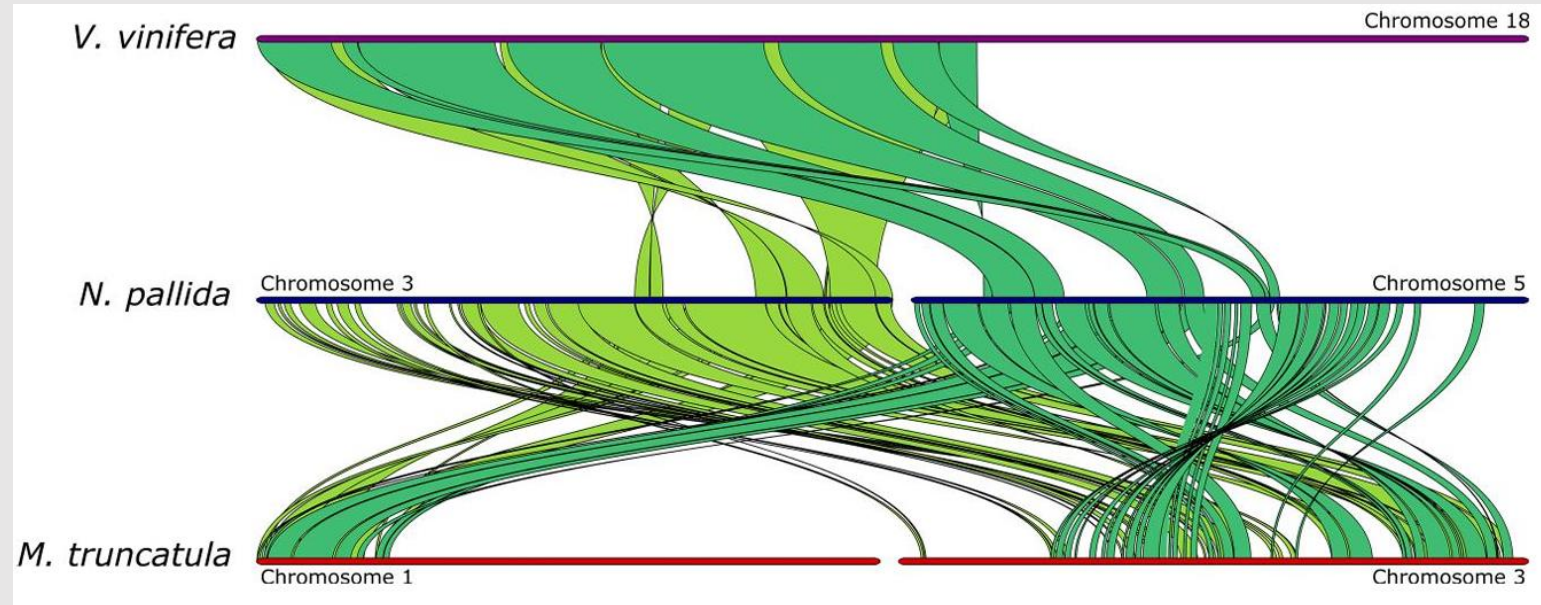


g_s : stomatal
conductance

- Genome sizes are correlated with latitude and environment (Souza et al. 2019; Bures et al. 2024).
- Genome sizes are correlated with better photosynthesis, plant growth and fitness (Simonin & Roddy, 2018; Faizullah et al. 2021).

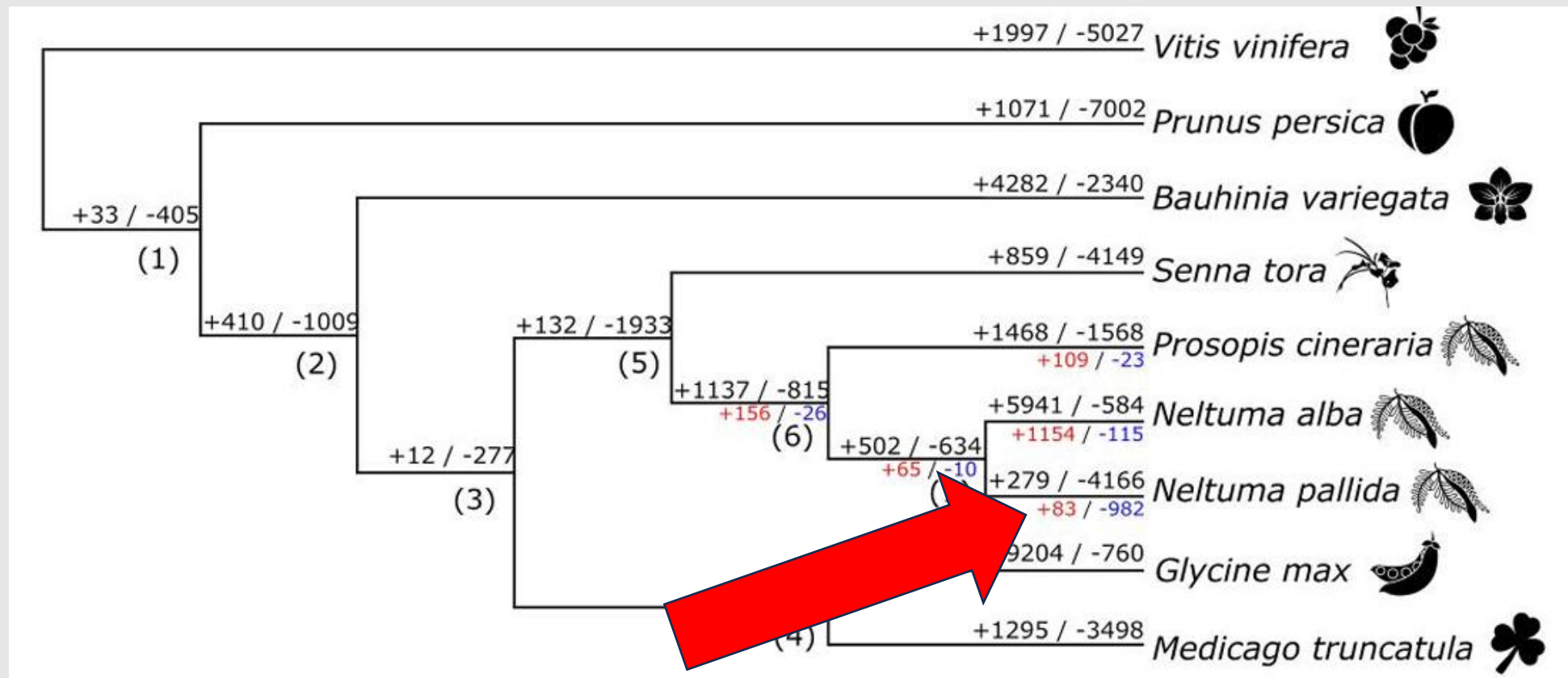
Genome downsizing of *N. pallida*

- Synteny analyses showed that the genome downsizing included also protein-coding regions.



Expanded against all odds

- In general, gene families are largely contracted or absent in *N. pallida*.
- However, some gene families were maintained or even expanded!



Seasonal plastic functions

Table 5. List of top five enriched GO terms for each category in significantly expanded gene families involving *N. pallida*.

Type	GO term	Description	P value
Biological process	GO:0009693 ^a	ethylene biosynthetic process	9.4E-13
	GO:0009727 ^a	detection of ethylene stimulus	5.3E-12
	GO:0010078	maintenance of root meristem identity	1.3E-11
	GO:0009056 ^a	catabolic process	7.5E-11
	GO:0080148	negative regulation of response to water deprivation	1.1E-10
MF	GO:0022821	solute:potassium antiporter activity	3.9E-14
	GO:0008061	chitin binding	5.4E-14
	GO:0030761 ^a	8-hydroxyquercetin 8-O-methyltransferase activity	1.1E-12
	GO:0050362	L-tryptophan:2-oxoglutarate aminotransferase activity	2.1E-12
	GO:0009815 ^a	1-aminocyclopropane-1-carboxylate oxidase activity	2.1E-12
Cellular component	GO:0009705	plant-type vacuole membrane	1.4E-6
	GO:0009543 ^a	chloroplast thylakoid lumen	1.5E-6
	GO:0016020	membrane	1.1E-5
	GO:0005618 ^a	cell wall	0.00047
	GO:0005773	vacuole	0.00126

^a GO terms also significantly enriched in gene families expanding only for the *N. pallida* branch.

Next steps

- What are the mechanisms underlying the genome downsizing in *N. pallida*?
- Are these gene expansions correlated with differential gene or protein expression? And replicable in common garden or field experiments?
- What else can the genome tell us more about *N. pallida* unique adaptation?
- How can this genomic resource be used for *N. pallida* deeper understanding and conservation?

Our collaborators



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Comprometidos por naturaleza

Cooking up the genome!



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- Eng. Sirley Bernabé, William Zeña
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- Dr. Carlos Arbizu and the INIA team
- Mg. Edgar Neyra



Thank you



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